PHYSICAL CHEMISTRY CUMULATIVE EXAMINATION
November 10, 1990

The First Law of Thermodynamics may be written:

$$dU = dq + dw$$

(a) Using the Second Law of Thermodynamics, write the expression for dU if a change of a closed system is brought about reversibly in the absence of any kind of work other than pV-work.

(b) The equation you just derived applies when the changes are reversible. However, dU is an exact differential and is independent of path. Therefore, dU(rev) = dU(irrev). This equation applies regardless of whether the change is brought about reversibly or irreversibly for any change in a closed system where there is no change of composition. Hence, this combination of First and Second Laws of Thermodynamics is called the fundamental equation. Using this equation, derive the equation for $(\partial U/\partial V)_T$.

(c) The following are the Maxwell relations:

$$\begin{aligned}
(\partial T/\partial V)_S &= -(\partial p/\partial s)_V \\
(\partial T/\partial P)_S &= (\partial V/\partial S)_P \\
(\partial p/\partial T)_V &= (\partial S/\partial V)_T \\
(\partial V/\partial T)_p &= -(\partial S/\partial p)_T
\end{aligned}$$

Using any of the Maxwell relations and the equation you derived for $(\partial U/\partial V)_T$, find what $(\partial U/\partial V)_T$ is for a perfect gas and for a van der Waals gas [i.e., $(p + an^2/V^2)(V-nb) = nRT$]. Explain in detail your interpretation of the differences in $(\partial U/dV)_T$.